17.1.2 Binomial Nomenclature

Carolus Linnaeus introduced the method of naming the animals with two names known as binomial nomenclature. The first name is called **genus** and the first letter of genus is denoted in capital and the second one is the **species** name denoted in small letter. The binomial names of some common animals are as follows.

Common name	Binomial name
Amoeba	*Amoeba proteus*
Hydra	*Hydra vulgaris*
Tapeworm	*Taenia solium*
Roundworm	*Ascaris lumbricoides*
Earthworm	*Lampito mauritii/ Perionyx excavatus*
Leech	*Hirudinaria granulosa*
Cockroach	*Periplaneta americana*
Snail	*Pila globosa*
Star fish	*Asterias rubens*
Frog	*Rana hexadactyla*
Wall lizard	*Podarcis muralis*
Crow	*Corvus splendens*
Peacock	*Pavo cristatus*
Dog	*Canis familiaris*
Cat	*Felis felis*
Tiger	*Panthera tigris*
Man	*Homo sapiens*

17.2 Invertebrata

17.2.1 Phylum Porifera (Pore bearers)

These are multicellular, non-motile aquatic organisms, commonly called as sponges. They exhibit **cellular grade of organization**. Body is perforated with many pores called **ostia**. Water enters into the body through ostia and leads to a **canal system**. It circulates water throughout the body and carries food, oxygen. The body wall contains **spicules**, which form the skeletal framework. Reproduction is by both asexual and sexual methods. e.g- *Euplectella, Sycon*.


Euplectella

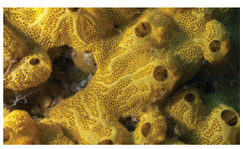

Sycon

Figure 17.3 Pore bearers

17.2.2 Phylum Coelenterata (Cnidaria)

Coelenterates are aquatic organisms, mostly marine and few fresh water forms. They are multicellular, radially symmetrical animals, with **tissue grade of organization**. Body wall is **diploblastic** with two layers. An outer ectoderm and inner endoderm are separated by non-cellular jelly like substance called **mesoglea**. It has a central gastrovascular cavity called **coelenteron** with mouth surrounded by short **tentacles**. The tentacles bear stinging cells called **cnidoblast** or **nematocyst**.

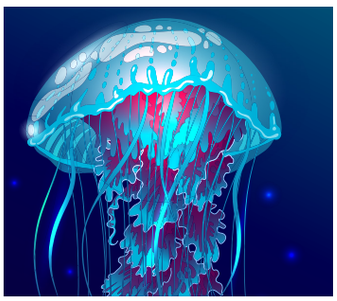


Figure 17.4 Jelly fish

Many coelenterates exhibit **polymorphism**, which is the variation in the structure and function of the individuals of the same species. They reproduce both asexually and sexually. e.g. Hydra, Jellyfish.

17.2.3 Phylum Platyhelminthes (Flat worms)

They are bilaterally symmetrical, triploblastic, **acoelomate** (without body cavity) animals. Most of them are **parasitic** in nature. **Suckers** and **hooks** help the animal to attach itself to the body of the host. Excretion occurs by specialized cells called **flame cells**. These worms are **hermaphrodites** having


Liver fluke


Tape worm

Figure 17.5 Flat worms

IX_SCI_EM_Unit-17_BIO.indd 202 12/7/2022 5:13:22 PM